

Assignment 4

Please finish the questions before Feb 10. You can use Stata to solve Question 1.

1. Using the VOLAT dataset for this question. The variable *rsp500* is the monthly return on the Standard & Poor's 500 stock index, at an annual rate. (This includes price changes as well as dividends. The variable *i3* is the return on three-month T-bill, and *pcip* is the percentage in industrial production; these are also at an annual rate.

(a) Consider the equation

$$rsp500_t = \beta_0 + \beta_1 pcip_t + \beta_2 i3_t + u_t.$$

What signs do you think β_1 and β_2 should have? Why?

(b) Estimate the previous equation by OLS and report the estimated regression model. Interpret the signs and magnitudes of the coefficients

(c) Which of the variables is statistically significant?

(d) Based on the empirical result in (b), use the t statistic to test for the null $H_0: \beta_2 = -1$ against $H_1: \beta_2 \neq -1$ at the 5% significance level.

(e) Does your finding from part (b) imply that the return on the S&P 500 is predictable? Explain.

(f) Are your conclusions for (b)-(d) reliable? Explain

2.

(a) Suppose the time series model below satisfies the Gauss-Markov assumptions. Explain the coefficients β_1 , β_2 , and β_3 .

$$GDP_t = \beta_0 + \beta_1 time + \beta_2 interest_rate_t + \beta_3 interest_rate_{t-1} + u_t$$

(b) Provide one time series example such that the y series' $Cov(y_t, y_{t-1}) > 0$.

(c) Provide one time series example such that the y series' $Cov(y_t, y_{t-1}) < 0$.